

Open the black box — from any model

Deck 1 read the model directly (coefficients, tree splits). But a random forest, a boosting model, a neural net — you *can't* read those.

Model-agnostic tools work on *any* fitted model and answer two questions:

- ▶ **How much** does a feature matter? → **Permutation Feature Importance (PFI)**.
- ▶ **How** does a feature act on the prediction? → **feature effects (PDP / ICE)**.

Running example continues: a **bike-sharing** random forest (deck 1's data).

Outline

Permutation feature importance

Feature effects: PDP & ICE

Feature interactions

How much does a feature matter?

Make a feature useless and watch the error rise.

Permutation feature importance: the idea

Make feature X_j useless by permuting its column (shuffle its values across rows): same marginal distribution, but its link to the target is broken. Then see how much the error rises:

$$\text{PFI}_j = \underbrace{\mathbb{E}[L(\hat{f}(X_j \text{ permuted}), Y)]}_{\text{error after}} - \underbrace{\mathbb{E}[L(\hat{f}(X), Y)]}_{\text{baseline error}}.$$

- ▶ Big rise $\Rightarrow$ the model leaned on X_j . Near zero $\Rightarrow$ it didn't.
- ▶ **Permute, don't drop** — the model is already trained; we can't remove a feature without refitting a *different* model.
- ▶ Repeat the shuffle several times and average (it's random).

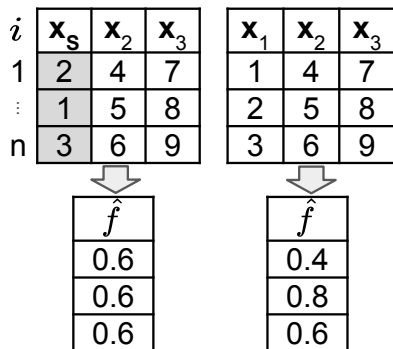
Permutation importance, step by step

i	$\mathbf{x}_s$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$
1	2	4	7	1	4	7
$\vdots$	1	5	8	2	5	8
n	3	6	9	3	6	9

Perturb (permute the feature) $\rightarrow$ **predict** on both versions $\rightarrow$ **aggregate** the change in loss, averaged over rows and repeats.

Source: LMU IML (CC BY 4.0).

Permutation importance, step by step



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1	2	4	7	1	4	7
$\vdots$	1	5	8	2	5	8
n	3	6	9	3	6	9

$L(\hat{f}, y)$
0.9
0.5
0.1

$L(\hat{f}, y)$
0.25
0.35
0.1

Perturb (permute the feature) → **predict** on both versions → **aggregate** the change in loss, averaged over rows and repeats.

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Permutation importance, step by step

i	$\mathbf{x}_s$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL
1	2	4	7	1	4	7	0.65
$\vdots$	1	5	8	2	5	8	0.15
n	3	6	9	3	6	9	0

$L(\hat{f}, y)$		$L(\hat{f}, y)$
0.9		0.25
0.5	-	0.35
0.1		0.1

Perturb (permute the feature) → **predict** on both versions → **aggregate** the change in loss, averaged over rows and repeats.

Source: LMU IML (CC BY 4.0).

Permutation importance, step by step

i	$\mathbf{x}_s$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL
1	2	4	7	1	4	7	0.65
$\vdots$	1	5	8	2	5	8	0.15
n	3	6	9	3	6	9	0

= 0.267

Perturb (permute the feature) $\rightarrow$ **predict** on both versions $\rightarrow$ **aggregate** the change in loss, averaged over rows and repeats.

Source: LMU IML (CC BY 4.0).

Permutation importance, step by step

1	i	$\mathbf{x}_s$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL	
	1	2	4	7	1	4	7	0.65	
	$\vdots$	1	5	8	2	5	8	0.15	$= 0.267$
	n	3	6	9	3	6	9	0	
m	i	$\mathbf{x}_s$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL	
	1	3	4	7	1	4	7	0.85	$\widehat{PFI}_s = \frac{1}{2} (0.267 + 0.4)$
	$\vdots$	2	5	8	2	5	8	0	
	n	1	6	9	3	6	9	0.35	$= 0.4$

Perturb (permute the feature) $\rightarrow$ **predict** on both versions $\rightarrow$ **aggregate** the change in loss, averaged over rows and repeats.

Source: LMU IML (CC BY 4.0).

PERMUTATION FEATURE IMPORTANCE

$$\mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}_{(k)}^S) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D})$$

1	i	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL	= 0.267
	1	2	4	7	1	4	7	0.65	
	$\vdots$	1	5	8	2	5	8	0.15	
	n	3	6	9	3	6	9	0	
m	i	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL	$\widehat{PFI}_S = \frac{1}{2} (0.267 + 0.4)$
	1	3	4	7	1	4	7	0.85	
	$\vdots$	2	5	8	2	5	8	0	
	n	1	6	9	3	6	9	0.35	

3. Aggregation:

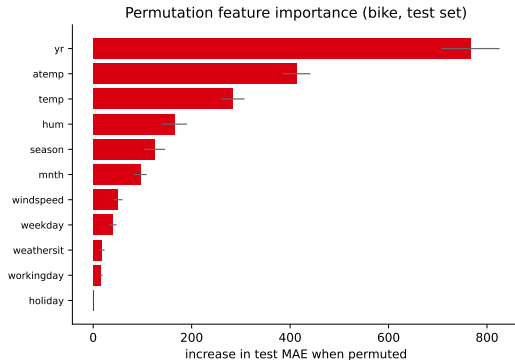
- Compute the loss for each observation in both data sets
- Take the difference of both losses ΔL for each observation
- Average this change in loss across all observations
- Repeat perturbation and average over multiple repetitions

PFI on the bike model

Permuting each feature on the **test set**, measured as the rise in MAE:

- ▶ `yr`, `atemp`/`temp` dominate. By hand: baseline MAE ≈ 455 ; shuffle `yr` $\rightarrow$ MAE ≈ 1225 ; $\text{PFI}(\text{yr}) \approx \mathbf{770}$.
- ▶ Calendar/weather minutiae barely matter.
- ▶ Error bars = variation over repeated shuffles.

Model-agnostic: same recipe would work on boosting, an SVM, a neural net.



Train or test? They answer different questions

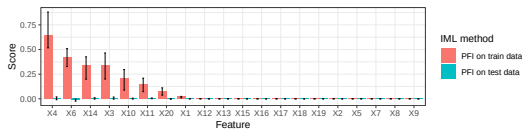
Predict-first: an xgboost model overfits noise. Which features does PFI flag — on train vs. on test?

Train or test? They answer different questions

Predict-first: an xgboost model overfits noise. Which features does PFI flag — on train vs. on test?

- ▶ **Train PFI** highlights whatever the model *overfit* to (spurious links).
- ▶ **Test PFI** shows what actually *generalizes* — here, nothing.

For “which features help the model generalize?”, compute PFI on **test** data.



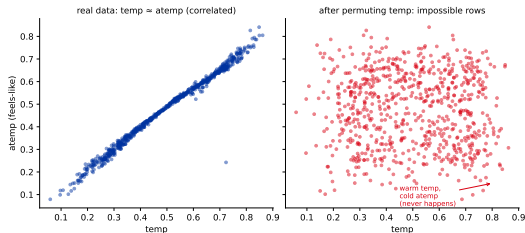
Illustrative example (not the bike model). Source: LMU IML (CC BY 4.0).

The correlation trap

Permuting a feature that is **correlated** with others creates **impossible rows** (e.g. a hot temp with a freezing atemp).

- ▶ The model is then scored on inputs it *never* sees in reality (extrapolation).
- ▶ $\Rightarrow$ PFI can **inflate** the importance of correlated features.

Note the flip: correlation *deflated* impurity importance (deck 1); here it *inflates* PFI. **One fix:** conditional variants (CFI) permute a feature *within groups of similar rows*, so the joint distribution stays realistic.



Bike: temp/atemp corr. 0.99; permuting temp makes impossible rows.

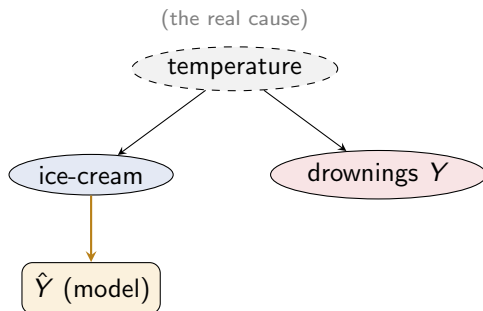
Important $\neq$ causal

Summer heat drives both **ice-cream sales** and **drownings**. A model predicting drownings from ice-cream sales finds ice cream **highly important**.

- ▶ PFI is right: the *model* relies on ice cream.
- ▶ But **banning ice cream won't stop drownings** — it isn't the cause.

Importance tells you what the **model uses**, not what *causes* the outcome.

In our data: *yr* is the #1 feature, yet the year doesn't *cause* rides — it proxies the service growing. (*Confounding — a different problem from the trap's impossible inputs.*)



Model uses the *proxy* (ice cream), not the cause (temperature).

From “how much” to “how”

Importance is one number; effects show the *shape* of a feature's influence.

Main effects and interactions: decomposing a prediction

Any model's prediction splits into pieces — a baseline, each feature acting *alone*, and features acting *together*:

$$\hat{f}(\mathbf{x}) = \underbrace{f_0}_{\text{baseline}} + \underbrace{\sum_j g_j(x_j)}_{\text{main effects}} + \underbrace{\sum_{j < k} g_{jk}(x_j, x_k)}_{\text{interactions}} + \dots$$

Main effect of x_j : how the prediction moves as x_j varies *alone* (averaging the rest). **The PDP (next slide) estimates exactly this.**

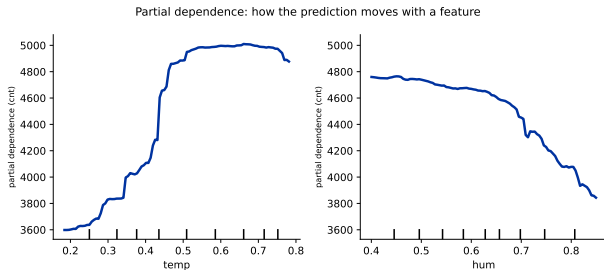
Interaction (x_j, x_k): the leftover when x_j 's effect *depends on* x_k . Zero for an **additive** model (e.g. linear with no product terms, deck 1).

Why care: if interactions are ≈ 0 the model is *additive* — its PDPs tell the whole story. Strong interactions mean a single feature's effect curve can **mislead**.

Partial dependence (PDP): the main-effect plot

Vary one feature across its range, average the model's prediction over all other rows, and plot the result — the feature's **main effect** g_j from the decomposition.

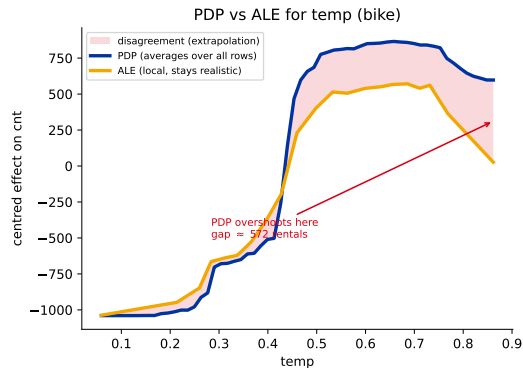
- temp: rentals climb with warmth, then plateau.
- hum: high humidity suppresses rentals.



PDP shares PFI's correlation trap — and ALE fixes it

To average over the other features, PDP also evaluates the model on **unrealistic combinations** when features are correlated (same extrapolation as the PFI correlation trap).

ALE (Accumulated Local Effects) fixes this: it averages *local* changes within small windows of the feature, staying inside the real data $\Rightarrow$ a correlation-robust effect curve.



Bike temp: PDP overshoots at the extremes; ALE stays realistic.

ICE: one curve per row — and what PDP hides

ICE draws one curve *per observation*;
the **PDP** is **their average**.

Predict-first: the PDP (orange) is **flat**
— does the feature not matter?

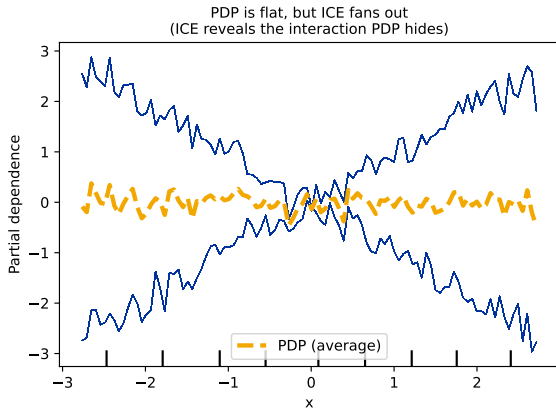
ICE: one curve per row — and what PDP hides

ICE draws one curve *per observation*;
the **PDP** is **their average**.

Predict-first: the PDP (orange) is **flat**
— does the feature not matter?

No. The ICE curves **fan out**: x 's effect
flips sign depending on another
feature — rising for half the rows,
falling for the other half, so the
opposing effects **cancel in the**
average. PDP hid that interaction; ICE
exposes it — exactly the kind of
interaction a *linear* model couldn't
represent (deck 1).

The visual test: *parallel* ICE curves = no interaction; curves that **fan out** = interaction. **Next:** see it
(2-D PDP) and *measure* it (H-statistic).



When one feature's effect depends on another

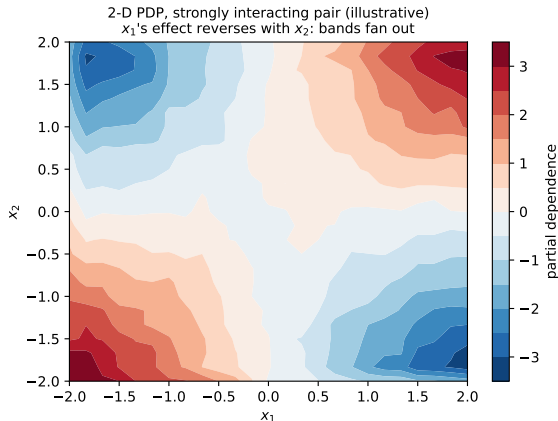
See it (the 2-D PDP), then put a *number* on it (the H-statistic).

Seeing an interaction: the 2-D PDP

Vary **two** features together and average over the rest — a partial-dependence **surface**.

- ▶ **No interaction** $\Rightarrow$ the bands are **parallel**: one feature's effect is the same at every value of the other.
- ▶ **Interaction** $\Rightarrow$ the bands **fan / bend**: here x_1 's effect **reverses** with the sign of x_2 (a clean $y = x_1x_2$ toy).

Illustrative on purpose — the bike model's *real* interactions are mild (we measure them next).



Measuring it: Friedman's H-statistic

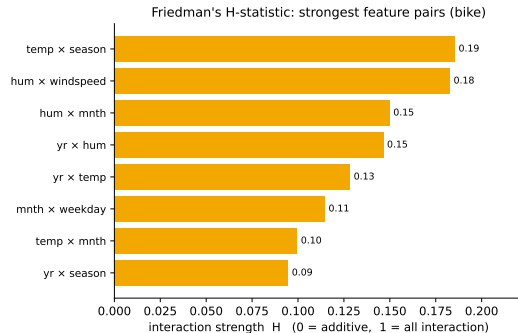
How much of a pair's effect is *interaction*, vs the two main effects simply added?

$$H_{jk}^2 = \frac{\sum_i [\text{PD}_{jk} - \text{PD}_j - \text{PD}_k]^2}{\sum_i [\text{PD}_{jk}]^2}$$

$\text{PD}_j, \text{PD}_{jk}$: the 1-D / 2-D PDPs from before; the joint PD_{jk} holds both mains, so the top line *strips* them $\rightarrow$ the interaction.

- ▶ $H = 0$: additive (joint = sum of mains).
- ▶ $H \rightarrow 1$: the effect is **almost all** interaction.

Costly ($O(n^2)$) and **noisy under correlation** — read it as a *ranking*, not an exact fraction. Not in `sklearn` (see the `artemis` package).



All $H < 0.2$: the bike model is **mostly main effects**; `temp×season` ($H=0.19$) leads, but modestly.

In sklearn

```
from sklearn.inspection import permutation_importance,
    PartialDependenceDisplay

# PFI on the TEST set -- works for ANY fitted model
r = permutation_importance(model, X_test, y_test, n_repeats=20,
                           scoring="neg_mean_absolute_error")
r.importances_mean      # rise in MAE per feature = importance

# feature effects: PDP (average) + ICE (one line per row)
PartialDependenceDisplay.from_estimator(model, X, ["temp", "hum"])
    # PDP
PartialDependenceDisplay.from_estimator(model, X, ["temp"], kind="
    both")  # PDP + ICE
PartialDependenceDisplay.from_estimator(model, X, [("temp", "season"
    )])  # 2-D PDP (interaction)
```

Recap

- ▶ **Model-agnostic** tools open any black box.
- ▶ **PFI** (*how much*): permute a feature, measure the error rise; compute it on **test** data; watch the **correlation trap**; and remember **important** $\neq$ **causal**.
- ▶ **PDP** (*how*): the **main-effect** shape. **ICE**: one curve per row — **PDP is their average**, and ICE reveals interactions the PDP hides.
- ▶ PDP shares PFI's extrapolation issue; **ALE** is the correlation-robust fix.
- ▶ **Main effect vs interaction**: prediction = baseline + main effects + interactions. The **2-D PDP** *shows* an interaction; **Friedman's H** *measures* it (bike: mostly main effects, temp $\times$ season strongest).

Next: SHAP — one method that explains a *single* prediction *and* rolls up to global importance & effects, with a rich set of plots.